2(a)

	gravitational potential energy / kJ	elastic potential energy / kJ	kinetic energy / kJ
top	24 (23.5)	0	0
half-way	12 (11.8)	2.6	9.2 (9.17) (accept 9.4)
bottom	0	24 (23.5)	0

Fig. 1.1

KE at bottom = 0 [A1] — **A1**

GPE (using mgh) [A1], — **A1**

KE at half-way and epe at bottom positions (using conservation of energy) [A1] — **A1**

2(b)

Let the spring constant of the elastic rope be k and unstretched length be x

At half-way position, $2.6 = \frac{1}{2}k(20-x)^2$

At bottom position, $23.5 = \frac{1}{2}k(40-x)^2$ [B1 for equations] — **B1**

$\frac{2.6}{23.5} = \frac{(20-x)^2}{(40-x)^2}$ — **M1**

$\sqrt{\frac{2.6}{23.5}} = \frac{20-x}{40-x}$

$x = 10\ m$ — **A0**

2(c)

Initially, when the bungee jumper jumps, her weight is greater than the tension in the elastic rope. When the tension increases until it is equal to the weight, she stops accelerating and reaches her highest velocity and kinetic energy.

$mg = ke$ where e is the extension when her speed is the highest — **C1**

$\frac{1}{2}kx^2 = \frac{1}{2}k(30)^2 = 23500$

$k = 52.2$ Nm^{-1} — **A1**

2(d)

$e = \frac{60 \times 9.81}{52.2}$ — **C1**

$= 11.3\ m$ — **A1**